

Shuo Guo

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Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)

Shenzhen, China

B.Sc. in Mathematics and Applied Mathematics

Sept 2025 – May 2029 (expected)

Experience

Research Assistant

May 2026 – Present

Working with PhD student in the research group of Prof. Ruoyu Sun, CUHK-Shenzhen. Currently involved in a project on optimization for deep learning, with a focus on optimizer benchmarking on NPUs.

Research Interests

- **Optimization Theory**
- **Deep learning from a mathematical perspective**

Skills

- **Convex Optimization**

By reading *Lectures on Convex Optimization* by Yurii Nesterov, I studied topics including convexity, gradient descent, and strongly convex complexity bounds. I'm especially interested in analyzing the convergence rate of GM, NM, AGM.

- **Data Science**

I attended Prof. Zhiqian Tom Luo's independent course, and read *Foundations of Data Science*. I learned SVD, random graphs and VC dimension.

- **Deep Learning**

My first RA experience was studying this field. I learned topics including FCN, backpropagation, optimization algorithms in DL, initialization, normalization layers and loss landscape. Also, I did some experiments on training GPTs and finding the best hyperparameters.

Also, I read *Mathematical Introduction to Deep Learning: Methods, Implementations, and Theory* and other theoretical articles in this field.

- **Stochastic Analysis,**

I enrolled in a course related to stochastic analysis, and learned almost all the important knowledge in this field.

- **Code**

I'm familiar with Python, Codex, GitHub Copilot, and other AI-assisted coding tools, and I can conduct basic programming for mathematical and statistical computing

Relevant Coursework

- Probability and Statistics
- Linear Algebra (Honor)
- Calculus I and II (Honor)
- Computer Science
- Stochastic Analysis